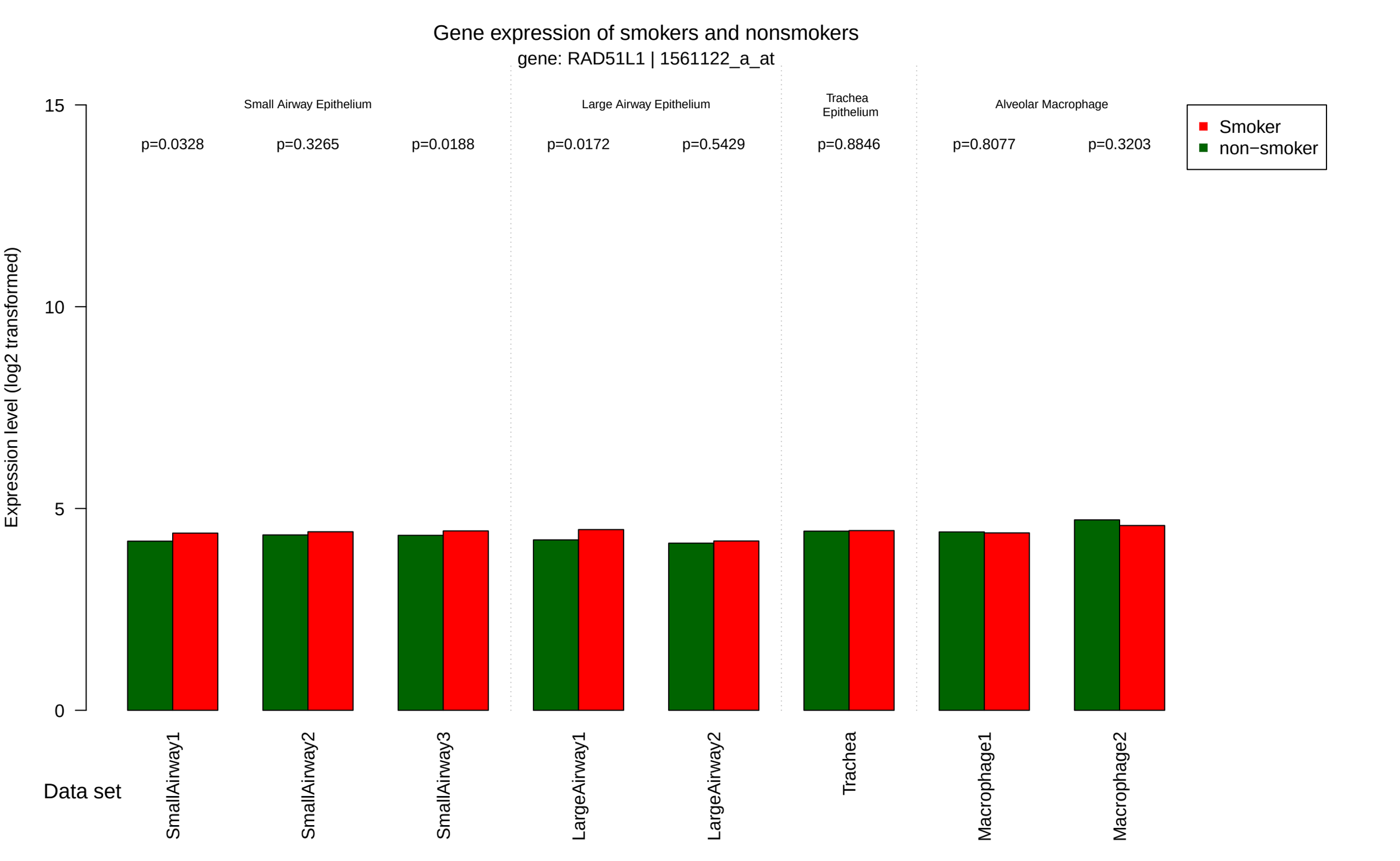
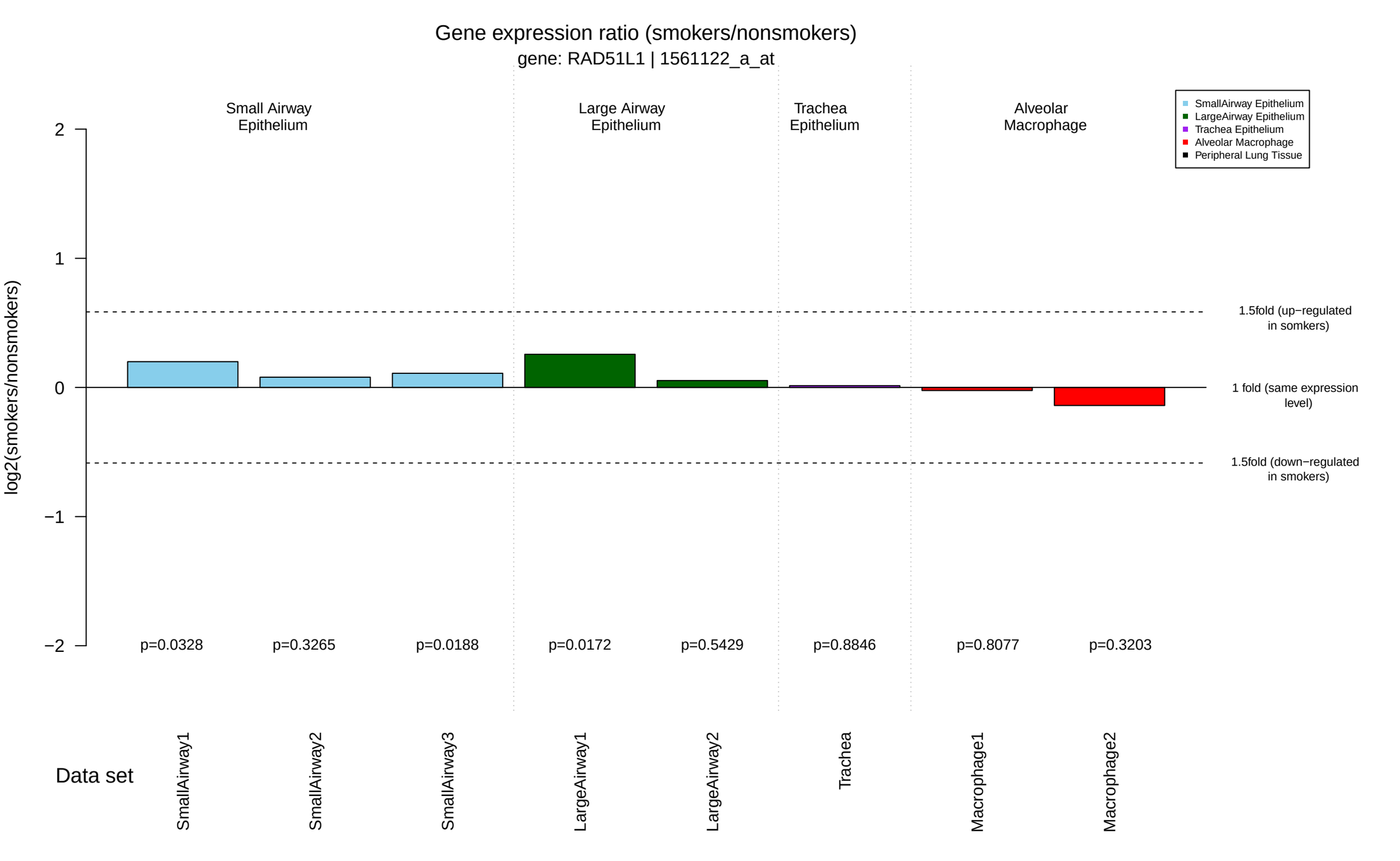
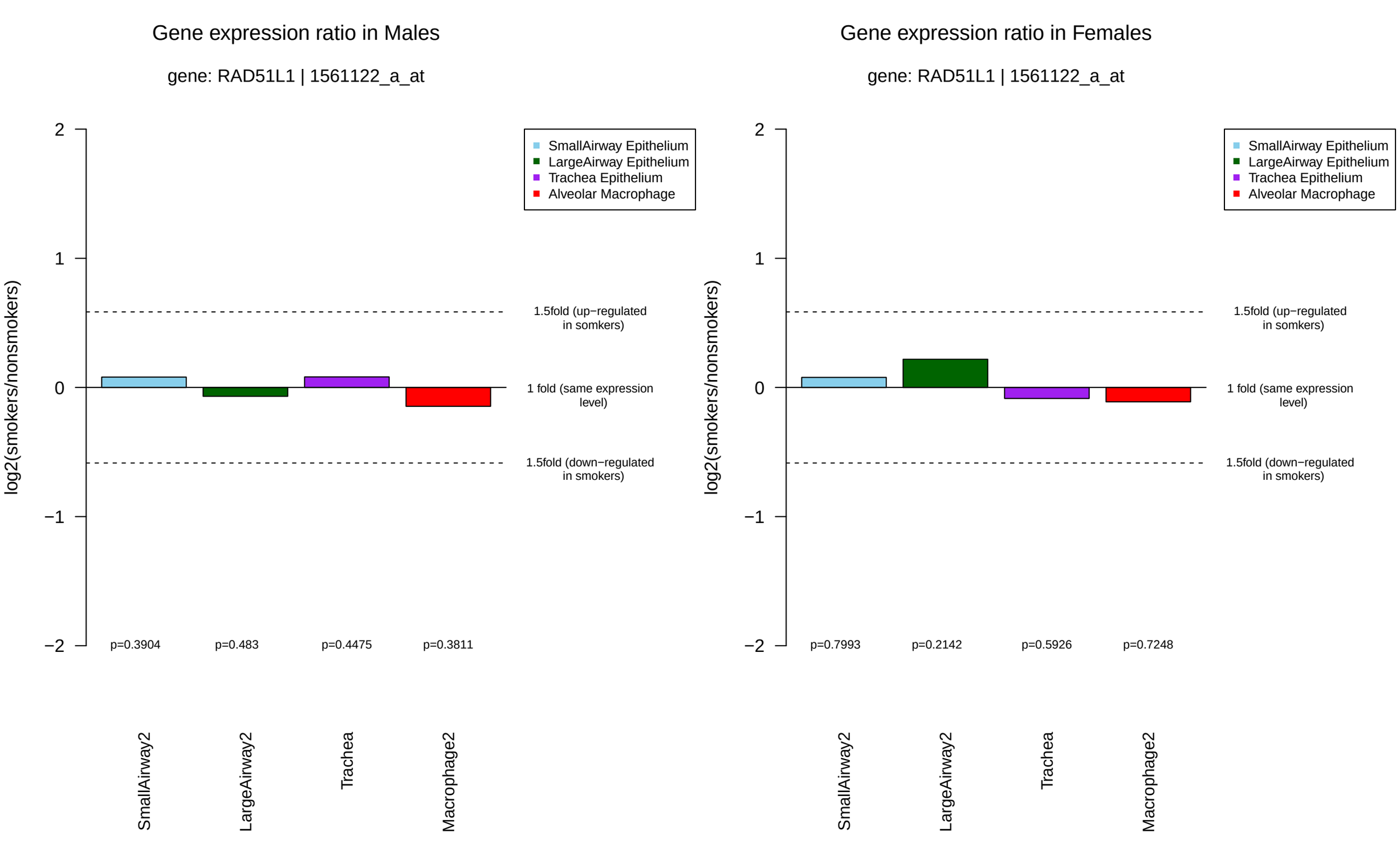
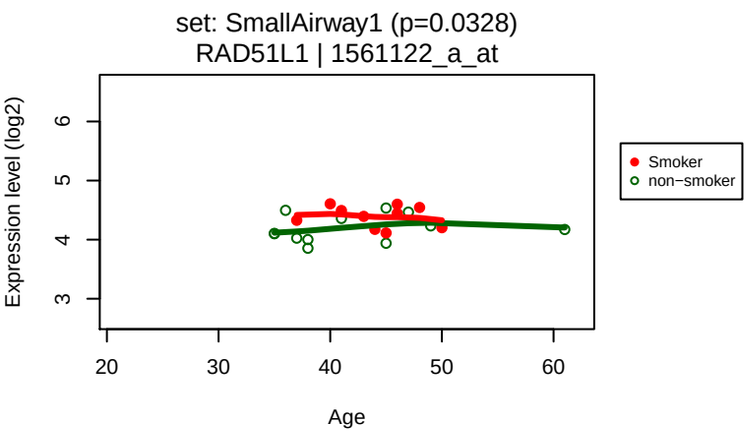
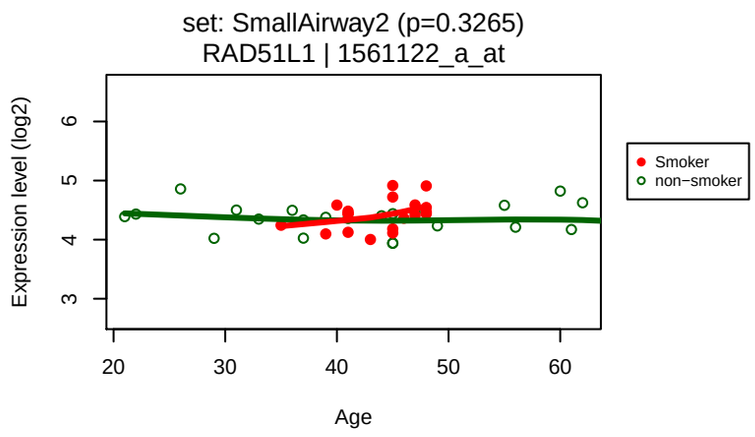
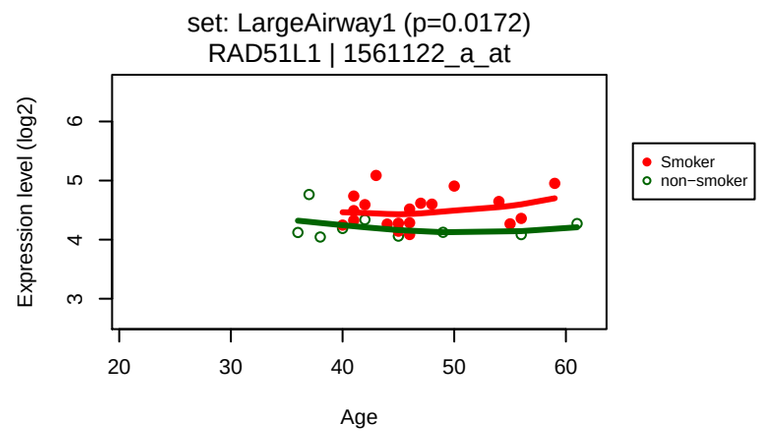

Gene expression of smokers and nonsmokers
gene: RAD51L1 | 1561122_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0328
p=0.3265
p=0.0188
p=0.0172
p=0.5429
p=0.8846
p=0.8077
p=0.3203
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

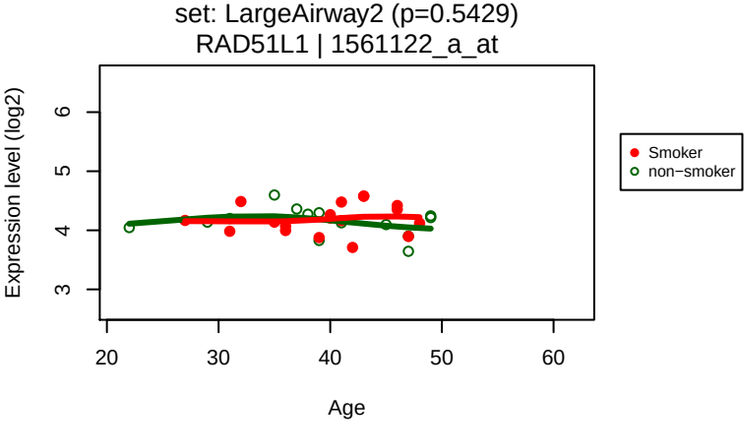

Gene expression ratio (smokers/nonsmokers)
gene: RAD51L1 | 1561122_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0328
p=0.3265
p=0.0188
p=0.0172
p=0.5429
p=0.8846
p=0.8077
p=0.3203
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

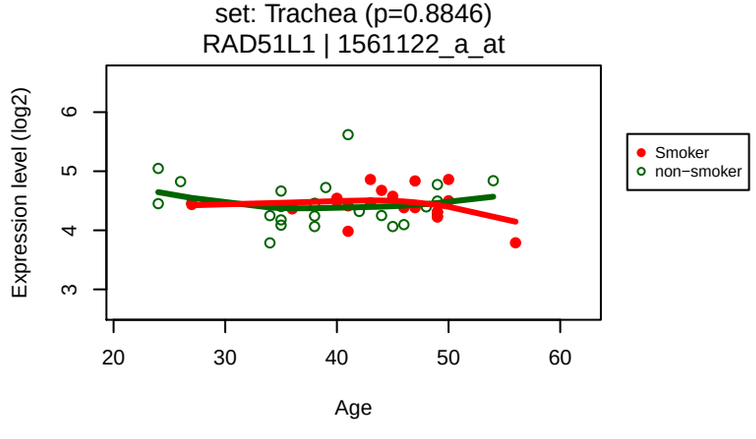

Gene expression ratio in Males
gene: RAD51L1 | 1561122_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3904
p=0.483
p=0.4475
p=0.3811
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAD51L1 | 1561122_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7993
p=0.2142
p=0.5926
p=0.7248
SmallAirway2
LargeAirway2
Trachea
Macrophage2

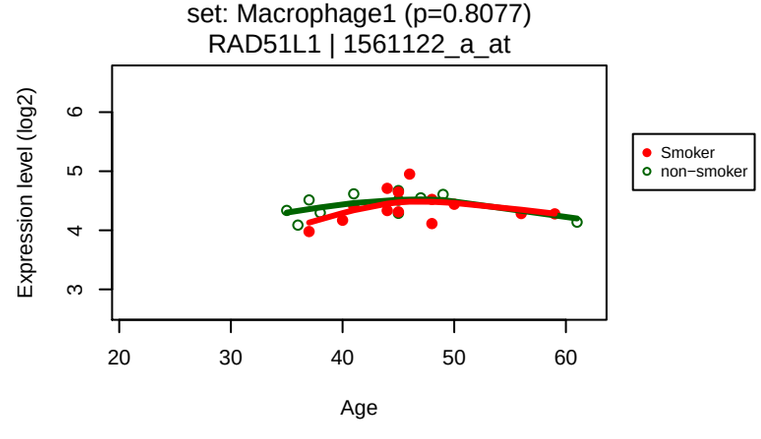

set: SmallAirway1 (p=0.0328)
RAD51L1 | 1561122_a_at
Expression level (log2)
Age
Smoker
non−smoker

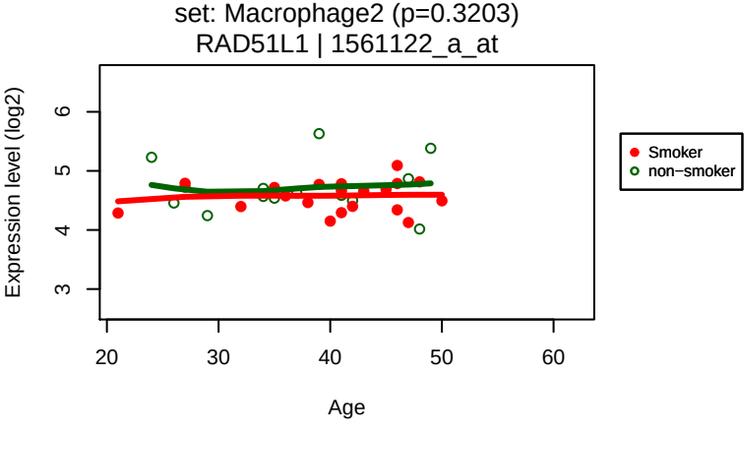

set: SmallAirway2 (p=0.3265)
RAD51L1 | 1561122_a_at
Expression level (log2)
Age
Smoker
non−smoker

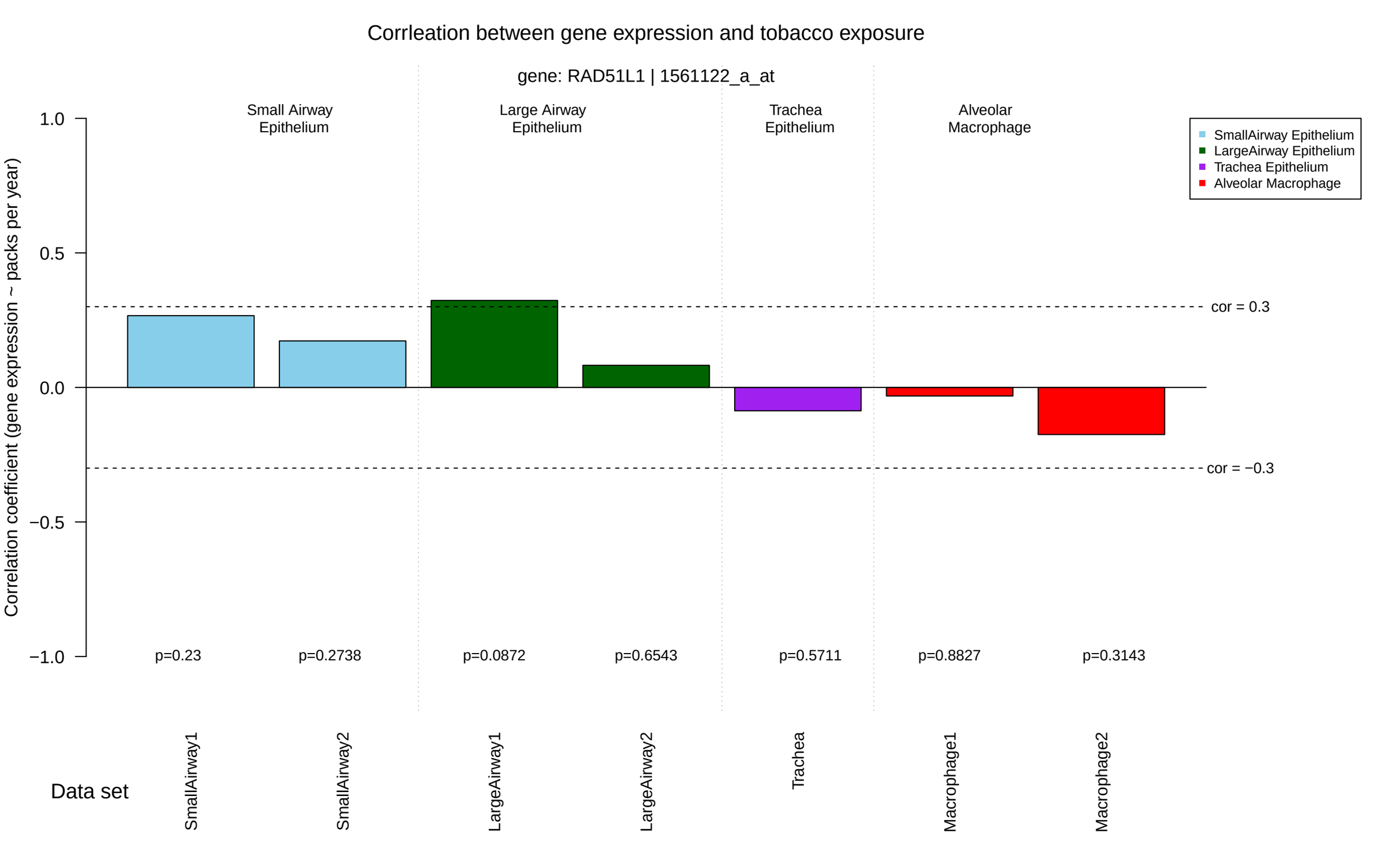

set: LargeAirway1 (p=0.0172)
RAD51L1 | 1561122_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5429)
RAD51L1 | 1561122_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8846)
RAD51L1 | 1561122_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8077)
RAD51L1 | 1561122_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3203)
RAD51L1 | 1561122_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAD51L1 | 1561122_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.23
p=0.2738
p=0.0872
p=0.6543
p=0.5711
p=0.8827
p=0.3143
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2